CODING–DECODING

Letter, Number, Sentence and Substitution Codes

OAVS TGT | Reasoning (Verbal) | Topic 5 of the syllabus list (Coding and Decoding)

Syllabus line (Reasoning, 30 Marks): “The Verbal Reasoning topics like Series, Ranks, Direction, Arrangement, **Coding, Decoding**, Analogy and Classification/Odd Pair, Syllogism and Statement Conclusion.”

Built from: OAVS TGT English 2019, English 2022, Mathematics 2024, Science 2024 and Social Studies 2023 (17 May 2023, Shift 2). Coding and Decoding are one question family (decoding is coding run backwards), so they are covered together in this one PDF. All 14 coding questions found in these papers are solved here.

What is inside

Ch.	Title	What you get
1	PYQ Analysis	Counts, patterns, priority, index of all 14 PYQs
2	Foundation	Alphabet table, opposite letters, coding vs decoding, 5-step method
3	Letter Coding	Shifts, alternating shifts, opposite letters, vowel-count questions, alphabet position
4	Number Codes	Letter-to-digit codes, word-value codes, conditional table codes
5	Word and Sentence Codes	Find a word’s code from sentences; substitution (artificial language)
6	Decoding and Expected Patterns	Reversing a code; patterns likely but not yet asked
7	Shortcuts, Traps and Scope	Speed methods; where related PYQs are covered
8	Practice Set	22 original questions with answer key
9	Quick Revision	One-page cheat sheet

SCOPE: Letter-cluster *relationships* (“A is related to B as C is related to ?”), odd-one-out and triad-pattern questions use the same letter shifts but belong to **Analogy** and **Classification**. Chapter 7 lists exactly which PYQs go where.

CHAPTER 1

PYQ Analysis

What the five papers tell us about Coding–Decoding

Paper	Q. numbers	Count	What was asked
English 2019	51, 52, 59, 60	4	+2 shift; alphabet position; opposite letters; letter-to-digit code
English 2022	44, 45	2	Alternating ±5 shift; substitution (artificial language)
Mathematics 2024	71, 89, 98	3	Next-letter vowel count (twice); sentence codes
Science 2024	75, 76, 77, 85	4	Sentence codes (twice); vowel count; word-value code
Social Studies 2023	30	1	Conditional number coding from a table
Total		**14**	**About 3 per paper; 7 of the 14 are from the two 2024 papers**

Pattern	PYQs	Priority
Letter shift +1 then count words with no vowel	3	★★★ Must
Sentence codes: find a word's code by common words	3	★★★ Must
Letter shifts: uniform (+2) and alternating (+5, −5)	2	★★★ Must
Opposite-letter (mirror) code	1	★★ High
Letter-to-digit code deduced from two words	1	★★ High
Word-value code (sum of letter positions × number of letters)	1	★★ High
Conditional coding with a table and rules	1	★★ High
Substitution: objects renamed	1	★★ High
Alphabet position (n-th letter to the left)	1	★★ High
Expected, not yet asked: reverse alphabet, increasing shifts, word-reversal codes, vowel/consonant rules	0	★ Expected

PYQ INSIGHT: The vowel-count template repeated: Maths 2024 Q71 (JET LAG RUM PIN), Maths 2024 Q98 (WON SEW HAT CUB) and Science 2024 Q77 (TRIP SPIN PINK TASK) are the same question with new words. Answers: Two, Two, Zero. A 10-second shortcut in 3.4 solves all three.

PYQ INSIGHT: Sentence codes appeared three times in 2024 (Maths Q89, Science Q75, Q76). Each asks for the code of one word and is solved by the 'word common to two sentences has the code common to both' method.

Index of the 14 PYQs

Paper / Q	Question in short	Ans	See
Eng 2019 Q51	NATION → PCVKQP; code of MARCONI	OCTEQPK	3.1
Eng 2019 Q52	14th letter to the left of the 21st letter	G	3.5
Eng 2019 Q59	RED → IVW, TEAR → GVZI; code of TRAVEL	GIZEVO	3.3
Eng 2019 Q60	PAPER → 74726, LEARN → 12469; code of NEPAL	92741	4.1
Eng 2022 Q44	BLACK → GGFXP, PINK → UDSF, BROWN → GMTRS; code of GREEN	LMJZS	3.2
Eng 2022 Q45	BUS called RICKSHAW … ; means of air transport	BOAT	5.2
Maths 2024 Q71	JET LAG RUM PIN: each letter to the next; words with no vowel	Two	3.4
Maths 2024 Q89	Sentence codes; code for 'good fun'	sm fu	5.1
Maths 2024 Q98	WON SEW HAT CUB: next letter; words with no vowel	Two	3.4
Science 2024 Q75	'go for it', 'it is beautiful'; code of 'it'	yt	5.1

Paper / Q	Question in short	Ans	See
Science 2024 Q76	Sentence codes; code for ‘and’	bq	5.1
Science 2024 Q77	TRIP SPIN PINK TASK: next letter; words with no vowel	Zero	3.4
Science 2024 Q85	FRESH → 280, FLOWER → 474; code of MIND	160	4.2
SST 2023 Q30	Table and conditions; code of MEANS	42387	4.3

CHAPTER 2

Foundation

Alphabet facts and the method behind every code

2.1 Alphabet table (memorise)

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26
26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1

Row 1: letters. Row 2: forward position. Row 3: backward position (position from Z). **Landmarks:** E = 5, J = 10, O = 15, T = 20, Y = 25. **Vowels:** A, E, I, O, U = 1, 5, 9, 15, 21. **Wrap-around:** after Z comes A (Y + 3 = 25 + 3 = 28, minus 26 = 2 = B); before A comes Z (A − 1 = 0, plus 26 = 26 = Z).

Opposite letters (positions add up to 27)

A–Z	B–Y	C–X	D–W	E–V	F–U	G–T	H–S	I–R	J–Q	K–P	L–O	M–N

2.2 Coding and decoding

Term	Meaning	What you do
Coding	A rule is given or hidden in examples; you **encode** a new word.	Find the rule from examples, apply it forward.
Decoding	A code is given; you find the **original** word.	Find the same rule, apply its **inverse** (+3 becomes −3).

2.3 The five-step method

Step	Do this
1	Write each given word and its code one under the other (letter for letter).
2	Convert to positions and compare position by position. Write the shift (+k, −k) under each pair.
3	Name the rule: same shift every time, alternating, increasing, opposite letters, digit mapping, or a value formula.
4	**Verify the rule on every example given** before using it. One failed example means a wrong rule.
5	Apply the rule to the question word. For a decoding question, apply the inverse. Check the answer against the options.

2.4 What you see → rule to test first

What you see	Test first
Code has the same number of letters as the word	Letter shift (same, alternating or progressive); opposite letters
Letters change but ‘wrongly’ in some places (+5 then −5)	Alternating shifts
Pairs like R–I, E–V, D–W (positions add to 27)	Opposite letters
Code is made of digits; one digit per letter	Letter-to-digit table from examples
Code is one number for a whole word	Sum of letter positions, then × letters (or other formula)
Whole sentences coded with 2-letter groups	Common-word method
Names of objects are swapped	Substitution: use only the new name of the real object

CHAPTER 3

Letter Coding

Shifts, mirrors and vowel-count questions: 7 PYQs

3.1 Uniform shift [★★★]

Rule: every letter moves the same number of places forward (+k) or backward (−k). Find k from the first pair.

PYQ Eng 2019 Q51 Same shift every time

Q. In a certain code, NATION is written as PCVKQP. How is MARCONI written?

Step 1. Compare N (14) → P (16): +2. A (1) → C (3): +2. T → V, I → K, O → Q, N → P: every letter is +2.
Step 2. Apply to MARCONI: M → O, A → C, R → T, C → E, O → Q, N → P, I → K.
Answer: OCTEQPK (Option A)

Letter	N	A	T	I	O	N
Position	14	1	20	9	15	14
Shift	+2	+2	+2	+2	+2	+2
Code	P	C	V	K	Q	P
Code position	16	3	22	11	17	16

SHORTCUT: Eliminate by the first letter: only OCTEQPK among the options starts with O (M + 2). Check the first and last letters, and you are done in 10 seconds.

3.2 Alternating shifts [★★★]

Rule: the shift changes in a cycle, such as +5, −5, +5, −5. Spot it when the same word position (1st, 3rd, 5th) always moves the same way.

PYQ Eng 2022 Q44 Alternating ±5

Q. BLACK is written as GGFXP, PINK as UDSF and BROWN as GMTRS. How is GREEN written?

Step 1. BLACK → GGFXP: B → G (+5), L → G (−5), A → F (+5), C → X (3 − 5 = −2, plus 26 = 24 = X, so −5), K → P (+5).
Step 2. Check PINK → UDSF (+5, −5, +5, −5) and BROWN → GMTRS (+5, −5, +5, −5, +5) ✓. Odd positions +5, even positions −5.
Step 3. GREEN: G +5 = L; R −5 = M; E +5 = J; E −5 = 0, plus 26 = Z; N +5 = S.
Answer: LMJZS (Option A)

Letter	B	L	A	C	K
Position	2	12	1	3	11
Shift	+5	-5	+5	-5	+5
Code	G	G	F	X	P
Code position	7	7	6	24	16

TRAP: Going below A wraps to Z: E − 5 = 0, which is Z (26), not a ‘zero’ letter. This single wrap-around is what separates LMJZS from the wrong options LMJYS, LMJJS, LMJXS.

3.3 Opposite-letter (mirror) code [★★]

Rule: each letter is replaced by the letter equally far from the other end (A ↔ Z, B ↔ Y …). Positions of a pair add up to 27.

PYQ Eng 2019 Q59 Opposite letters

Q. In a certain code RED is written as IVW and TEAR as GVZI. How is TRAVEL written?

Step 1. R (18) → I (9): 18 + 9 = 27. E (5) → V (22): 5 + 22 = 27. D (4) → W (23): 27. Every pair adds up to 27: opposite letters.
Step 2. TRAVEL: T ↔ G, R ↔ I, A ↔ Z, V ↔ E, E ↔ V, L ↔ O.
Answer: GIZEVO (Option B)

SHORTCUT: Test in 5 seconds: add the positions of one letter and its code. If the total is 27, it is a mirror code; every other letter will do the same.

3.4 Next letter, then count vowels [★★★ three PYQs]

Template: 'If each letter of the words is changed to the next letter, how many words will have no vowel?' The words are 3 or 4 letters long.

Shift	A new letter is a vowel only if the original letter is…	Memory aid
+1 (next letter)	**Z, D, H, N, T** (Z → A, D → E, H → I, N → O, T → U)	ZD HNT
−1 (previous letter)	**B, F, J, P, V** (B → A, F → E, J → I, P → O, V → U)	BF JPV

PYQ	Words	Letters from Z, D, H, N, T	Words with a vowel after the shift	No vowel
Maths 2024 Q71	JET, LAG, RUM, PIN	JET: T; PIN: N	KFU, QJO	**Two** (LAG → MBH, RUM → SVN)
Maths 2024 Q98	WON, SEW, HAT, CUB	WON: N; HAT: H, T	XPO, IBU	**Two** (SEW → TFX, CUB → DVC)
Science 2024 Q77	TRIP, SPIN, PINK, TASK	T, N, N, T in every word	USJQ, TQJO, QJOL, UBTL	**Zero**

PYQ Maths 2024 Q71 Step-by-step

Q. JET, LAG, RUM, PIN: each letter is changed to the next letter. How many words will have no vowel?

Step 1. JET → K, F, U = KFU: contains U (vowel).
Step 2. LAG → M, B, H = MBH: no vowel.
Step 3. RUM → S, V, N = SVN: no vowel.
Step 4. PIN → Q, J, O = QJO: contains O (vowel).
Step 5. Words with no vowel: MBH and SVN = 2.
Answer: Two (Option B)

SHORTCUT: Shortcut: scan each original word for Z, D, H, N or T. If it has one of them, the new word has a vowel. Count the words that have none of these letters. Remember that vowels in the *original* word turn into consonants (A → B, E → F, I → J, O → P, U → V).

3.5 Alphabet position [★★]

Phrase	Rule	Example
n-th letter to the right of letter at position p	p + n	5th to the right of J (10) = O (15)
n-th letter to the left of letter at position p	p − n	14th to the left of U (21) = G (7)
n-th letter from the right end	27 − n	15th from the right end = 12 = L
Letters between two letters	difference − 1	Between C (3) and J (10): 6 letters

PYQ Eng 2019 Q52 Alphabet position

Q. What would be the 14th letter to the left of the twenty-first letter of the English alphabet?

Step 1. The 21st letter is U.
Step 2. 14th to the left means position 21 − 14 = 7.
Step 3. Position 7 is G.
Answer: G (Option B)

CHAPTER 4

Number Codes

Letters coded as digits or numbers: 3 PYQs

4.1 Letter-to-digit code deduced from examples [★★]

Method: write each word's letters over its digits, extract every letter's digit, then check that repeated letters give repeated digits.

PYQ Eng 2019 Q60 Digit per letter

Q. If PAPER is coded as 74726 and LEARN as 12469, how is NEPAL coded?

Step 1. PAPER = 7 4 7 2 6: P = 7, A = 4, E = 2, R = 6 (the repeated P gives 7 twice ✓).
Step 2. LEARN = 1 2 4 6 9: L = 1, E = 2, A = 4, R = 6, N = 9 ✓ (E, A, R agree with PAPER).
Step 3. NEPAL: N = 9, E = 2, P = 7, A = 4, L = 1.
Answer: 92741 (Option D)

Letter	P	A	E	R	L	N
Digit	7	4	2	6	1	9

4.2 Word-value codes (sum of positions) [★★]

Method: find the sum of letter positions, then test simple formulas against both examples: sum × number of letters, sum + number of letters, 2 × sum.

PYQ Science 2024 Q85 Sum × number of letters

Q. FRESH is coded as 280 and FLOWER as 474. What is the code for MIND?

Step 1. FRESH: F 6 + R 18 + E 5 + S 19 + H 8 = 56. Five letters: 56 × 5 = 280 ✓.
Step 2. FLOWER: F 6 + L 12 + O 15 + W 23 + E 5 + R 18 = 79. Six letters: 79 × 6 = 474 ✓.
Step 3. MIND: M 13 + I 9 + N 14 + D 4 = 40. Four letters: 40 × 4 = 160.
Answer: 160 (Option 3)

SHORTCUT: Test your guess on both examples before applying. Here 280 ÷ 5 = 56 and 474 ÷ 6 = 79: both whole numbers is the hint that the code is 'sum × letters'.

4.3 Conditional coding with a table [★★]

Method: (1) write the basic code from the table; (2) check each condition in the order given; (3) apply the condition only if the word meets it; (4) some conditions change the earlier result.

PYQ SST 2023 Q30 Letters coded by a table and two conditions

Q. Letters S, M, A, E, N have codes 4, 7, 3, 2, 8. (i) If the first and last letters are consonants, interchange their codes. (ii) If the fourth letter is a vowel, code it as 9. Find the code for MEANS.

Step 1. Basic code: M 7, E 2, A 3, N 8, S 4 → 7 2 3 8 4.
Step 2. Condition (i): the first letter M and the last letter S are both consonants, so interchange their codes: 4 2 3 8 7.
Step 3. Condition (ii): the fourth letter is N, a consonant, so no change.
Answer: 42387 (Option 4)

Letter	M	E	A	N	S
Basic code	7	2	3	8	4
After (i)	4	2	3	8	7
After (ii)	4	2	3	8	7 (no change)

TRAP: A condition can be false. In MEANS the fourth letter is N (a consonant), so 'code the vowel as 9' does **not** apply. Option 2 (72384) is what you get if you ignore condition (i).

CHAPTER 5

Word and Sentence Codes

Find a word's code from several sentences, and substitution: 4 PYQs

5.1 The common-word method [★★★ three PYQs]

Step	Do this
1	Find a word that appears in **two** sentences. Its code is the code that appears in **both** codes.
2	Remove those codes. Repeat with other pairs of sentences.
3	A word that occurs in only one sentence takes the **last unexplained code** of that sentence.
4	Match the question word with its code and check the options.

PYQ Science 2024 Q75 Two sentences

Q. 'go for it' is coded 'az bx yt' and 'it is beautiful' is coded 'yt km fk'. How is 'it' coded?

Step 1. 'it' appears in both sentences.
Step 2. The code common to both: 'yt' appears in both codes (az bx yt and yt km fk).
Step 3. So 'it' = yt.
Answer: yt (Option 4)

PYQ Science 2024 Q76 Three sentences

Q. 'crunching those numbers' = 'dy ph mt'; 'analyzing sales numbers' = 'lp ph st'; 'crunching and analyzing' = 'st bq dy'. What is the code for 'and'?

Step 1. 'numbers' is in sentences 1 and 2; the common code is ph. So numbers = ph.
Step 2. 'crunching' is in sentences 1 and 3; the common code is dy. 'analyzing' is in sentences 2 and 3; the common code is st.
Step 3. In sentence 3 ('crunching and analyzing' = st bq dy), the unexplained code is bq: 'and' = bq.
Answer: bq (Option 4)

Maths 2024 Q89

'is tomorrow a holiday' = kb wo vg hv; 'holiday is good fun' = fu kb sm vg; 'good health is essential' = vg mc rp sm; 'health and wealth together' = cy zt gd rp (all codes two letters). What is the code for 'good fun'?

Word	In sentences	Common code	Code
is	1, 2, 3	vg appears in all three	vg
holiday	1, 2	kb and vg are common; vg is 'is', so kb	kb
good	2, 3	sm and vg are common; vg is 'is', so sm	sm
health	3, 4	rp appears in both	rp
fun	2 only	sentence 2: fu is left (kb, sm, vg are taken)	fu

RESULT: sm fu (Option C): 'good fun' = sm fu. **Option check:** the correct option must contain sm; the options with kb or vg cannot be right because those are 'holiday' and 'is'.

5.2 Substitution (artificial language) [★★]

Rule: the question names an object, and the answer is the **new name** given to it. Do **not** continue down the chain: each renaming applies once.

PYQ Eng 2022 Q45 Objects renamed

Q. If BUS is called RICKSHAW, RICKSHAW is called CAR, CAR is called MOTORCYCLE, MOTORCYCLE is called AEROPLANE, and AEROPLANE is called BOAT, what is the means of air transport?

Step 1. The real means of air transport is the aeroplane.
Step 2. AEROPLANE is called BOAT.
Step 3. So the answer is BOAT.
Answer: BOAT (Option A)

TRAP: Do not chain: BUS is called RICKSHAW, but that does not make BUS also a CAR. The answer is 'BOAT', not 'AEROPLANE' (the real name) and not something further along the chain.

CHAPTER 6

Decoding and Expected Patterns

Reversing a code and patterns likely in future papers

6.1 Decoding: apply the inverse

Coding rule	Inverse for decoding	Example
Shift +k	Shift −k	KHOOR (+3) → HELLO
Shift −k	Shift +k	ADC (−1) → BED
Alternating +5, −5	Alternating −5, +5	GGFXP → BLACK
Opposite letters	Opposite letters (self-inverse)	XLWV → CODE
Letter-to-digit table	Read the table backwards	92741 → NEPAL

SHORTCUT: A mirror (opposite-letter) code is its own inverse: coding and decoding use the same step. For every other shift, remember to **flip the sign**.

6.2 Patterns likely but not yet asked

Pattern	Rule	Worked example
Progressive shift	Shifts +1, +2, +3 ... by position	WORD → W+1 X, O+2 Q, R+3 U, D+4 H = XQUH
Reverse alphabet numbering	A = 26, B = 25 ...; value = 27 − normal position	M (13) = 14

Pattern	Rule	Worked example
Reversing the word	Letters written backwards, sometimes then shifted	FORT → TROF; PORTS → STROP then +1 each
Vowel / consonant rule	Vowels get one rule, consonants another	Vowels +1, consonants −1
Position-sum with a bonus	Sum of positions plus or times a number	CAT = 3 + 1 + 20 = 24; × 3 letters = 72

EXPECTED: These are predictions from the common question bank, not from your five papers. Spend time on Chapters 3 to 5 first; use this table for a last revision.

CHAPTER 7

Shortcuts, Traps and Scope

Marks saved, marks lost, and where related PYQs belong

- **Check the first and last letter first**: it removes most options in seconds (OCTEQPK, LMJZS).
- **Add positions of a pair**: a total of 27 means a mirror code.
- **Same word in two sentences**: its code is the common code (all three 2024 sentence-code questions).
- **Vowel-count template**: shift +1, look for Z, D, H, N, T; shift −1, look for B, F, J, P, V.
- **Verify on every example** before applying a rule: an alternating code looks like a uniform code on the first pair.
- **Wrap-around**: add 26 when you go below 1; subtract 26 when you go above 26.
- **Value codes**: test 'sum × letters' first; check that both examples divide evenly.

Trap	How to avoid it
Using only the first letter pair to find the rule	Compare all letters; BLACK looks like +5 until the second letter (−5).
Forgetting the wrap-around	E − 5 is Z (26 − 0), Y + 3 is B.
Counting the original vowels in the vowel-count question	Check the **new** letters, not the old ones.
Substitution: following the chain	Use only one renaming (BOAT, not AEROPLANE).
Conditional coding: applying every condition	A condition applies only if the word meets it.
Decoding with the wrong sign	Shift +3 is decoded with −3.
Number code applied to the wrong letter	Write the letter table (7 4 2 6 1 9) before decoding.

Scope: PYQs that look like coding but belong to other topics

PYQ	Why not here
Maths 2024 Q92, Q95; Science 2024 Q74, Q82, Q90; Eng 2022 Q52; SST 2023 Q10	'A is related to B as C is related to ?' (letter-cluster, number and word analogies): Analogy
Maths 2024 Q90, Q100; Science 2024 Q99	'Select the triad that follows the same pattern': Analogy
Maths 2024 Q72, Q85; Eng 2022 Q51, Q58, Q59; Science 2024 Q72, Q87, Q89; SST 2023 Q5, Q29	Odd letter-cluster or odd word-pair: Classification / Odd Pair
Maths 2024 Q94; Science 2024 Q92	Symbols for family relations ('A + B means A is the son of B'): blood relations are not in the listed syllabus
Eng 2022 Q57; Science 2024 Q81, Q94	Equation signs and digit operations: not in the listed syllabus

CHAPTER 8

Practice Set

22 original questions; answers on the next page

Q	Question
1	If CAT is coded as ECV, how is DOG coded?
2	In a code, LOVE is written as ORYH. How is MIND written?
3	If SCHOOL is written as RBGNNK, how is TEACHER written?
4	In a code, WORD is written as XQUH. How is PLAY written in that code?
5	If FAME is written as UZNV, how is LOVE written?
6	If BEAM is written as DCCK, how is HOME written?
7	BED, TOP, SKY, NIP: each letter is changed to the previous letter of the alphabet. How many words will have no vowel?
8	What is the 6th letter to the right of the 15th letter from the right end of the English alphabet?
9	If CAT is coded as 72, how is DOG coded (the same rule)?
10	If TEAM is coded as 5231 and MATE as 1352, how is MEAT coded?
11	Letters B, A, N, D, E have codes 4, 7, 2, 9, 5. (i) If the first and last letters are consonants, interchange their codes. (ii) If the second letter is a vowel, code it as 1. What is the code for BAND?
12	‘sun is bright’ = ‘ka mo pi’; ‘bright star shines’ = ‘pi lu ze’; ‘sun and star’ = ‘ka ri lu’. What is the code for ‘and’?
13	‘red rose blooms’ = ‘ab cd ef’; ‘rose petals fall’ = ‘cd gh ij’; ‘blooms and fall’ = ‘ef kl ij’. What is the code for ‘petals’?
14	If sky is called water, water is called road, road is called tree, tree is called book, book is called cloud, where do fish swim?
15	If white is called blue, blue is called red, red is called yellow, yellow is called green, what is the colour of human blood?
16	TAP, FOG, NUN, EAR: each letter is changed to the next letter. How many words will have no vowel?
17	If MOTHER is written as NPUIFS, how is FATHER written?
18	If DELHI is coded as 73541 and CALCUTTA as 82589662, how is CALICUT coded?
19	If the alphabet is numbered in reverse order (Z = 1, Y = 2 …), what is the number of M?
20	KHOOR is the code obtained by moving each letter 3 places forward. What is the original word?
21	In a code each letter is replaced by its opposite letter (A ↔ Z). What does XLWV stand for?
22	If ROSE is written as TQUG, which word is written as PGYU?

Answer key with logic

Q	Answer	Logic
1	FQI	+2 each letter: D → F, O → Q, G → I
2	PLQG	+3: M → P, I → L, N → Q, D → G
3	SDZBGDQ	−1: T → S, E → D, A → Z, C → B, H → G, E → D, R → Q
4	QNDC	+1, +2, +3, +4: P → Q, L → N, A → D, Y → C (25 + 4 = 29 − 26 = 3)
5	OLEV	Opposite letters: L ↔ O, O ↔ L, V ↔ E, E ↔ V
6	JMOC	+2, −2, +2, −2: H → J, O → M, M → O, E → C
7	One	BED → ADC (A), TOP → SNO (O), SKY → RJX (none), NIP → MHO (O): only SKY has no vowel
8	R	15th from the right end = 27 − 15 = 12 = L; 12 + 6 = 18 = R
9	78	CAT: 3 + 1 + 20 = 24 × 3 = 72 ✓; DOG: 4 + 15 + 7 = 26 × 3 = 78
10	1235	T = 5, E = 2, A = 3, M = 1; MEAT = 1 2 3 5
11	9124	Basic: B 4, A 7, N 2, D 9 → 4 7 2 9. B and D are consonants: swap first and last → 9 7 2 4. Second letter A is a vowel → 1: 9 1 2 4
12	ri	sun = ka (1, 3); star = lu (2, 3); bright = pi (1, 2); ‘and’ = the remaining code in sentence 3: ri
13	gh	rose = cd; fall = ij; blooms = ef; red = ab; ‘petals’ = gh (left in sentence 2)
14	Road	Fish swim in water; water is called road
15	Yellow	Blood is red; red is called yellow
16	Two	TAP → UBQ (U), FOG → GPH (none), NUN → OVO (O), EAR → FBS (none): FOG and EAR
17	GBUIFS	+1 each letter
18	8251896	C 8, A 2, L 5, I 1, C 8, U 9, T 6 (from DELHI and CALCUTTA)
19	14	27 − 13 = 14
20	HELLO	Decode by −3: K → H, H → E, O → L, O → L, R → O
21	CODE	X ↔ C, L ↔ O, W ↔ D, V ↔ E
22	NEWS	ROSE → TQUG is +2, so decode PGYU with −2: P → N, G → E, Y → W, U → S

CHAPTER 9

Quick Revision

One page for the last day

Question type	Method	PYQ check
Uniform shift	Compare the first pair; apply to every letter	NATION → PCVKQP (+2); MARCONI → OCTEQPK
Alternating shift	Odd positions one way, even positions the other	BLACK → GGFXP (+5, −5); GREEN → LMJZS
Opposite letters	Pair positions add up to 27	RED → IVW; TRAVEL → GIZEVO
Next-letter vowel count	Shift +1: look for Z, D, H, N, T. Shift −1: look for B, F, J, P, V	Two, Two, Zero in 2024
Alphabet position	Right = +n, left = −n, from the right end = 27 − n	14th to the left of U = G
Letter-to-digit code	Extract each letter's digit from the examples	PAPER, LEARN → NEPAL = 92741
Word value	Sum of positions × number of letters	FRESH 56 × 5 = 280; MIND 40 × 4 = 160
Conditional table code	Basic code, then apply only the conditions that hold	MEANS = 42387
Sentence codes	Common word ↔ common code	'it' = yt; 'and' = bq; 'good fun' = sm fu
Substitution	Answer is the new name of the real object; no chaining	Air transport → aeroplane → BOAT
Decoding	Apply the inverse (+3 → −3); mirror codes are self-inverse	KHOOR → HELLO; XLWV → CODE

ROUTINE: Exam-day routine: write the example word and code in two rows, find the shift for every letter (not only the first), verify on all examples, then apply. For sentence codes: find the word that repeats. 30 seconds per question.